Industrial Internship Report on

” Predictive Maintenance & RAM Pricing Intelligence System”

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time.

My project focused on Predictive Maintenance and RAM Pricing Intelligence Data Engineering/Machine Learning Solutions, leveraging data processing, feature engineering, and predictive modelling to solve real-world industrial optimization problems.

This internship provided an excellent opportunity to gain exposure to industrial challenges and design and implement end-to-end solutions. Overall, it was a rewarding experience.

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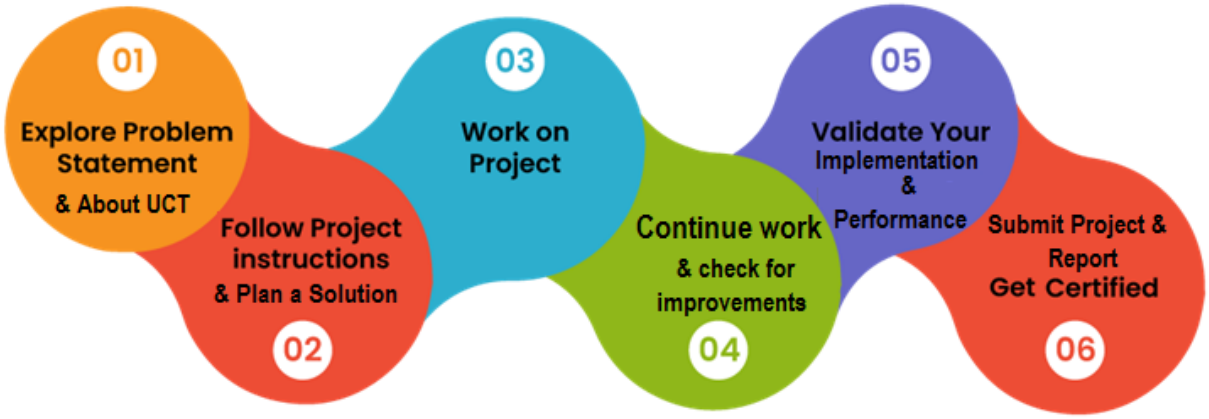
1 Preface

Over the course of 6 weeks, the work progressed through requirement analysis, data cleaning, exploratory data analysis (EDA), model training, algorithm optimization, evaluation, and documentation.

Practical internships bridge academic knowledge and industrial implementation. They provide exposure to production constraints, practical performance metrics, and realistic data pipelines.

The project focused on predictive analytics and machine learning applications to analyze data trends, extract meaningful features, and build accurate predictive models for industrial decision-making.

Facilitated by upSkill Campus and UniConverge Technologies, the internship followed a structured 6-step lifecycle:



Sincere gratitude to the mentors and coordinators at UniConverge Technologies, upSkill Campus, and The IoT Academy for their guidance throughout the internship program.

Message to Peers & Juniors: Engage actively with practical datasets, focus on model interpretability and validation, and build strong foundational skills in data pre-processing and pipeline design.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

Established in 2013, UniConverge Technologies Pvt Ltd (UCT) operates in the Digital Transformation domain, providing industrial solutions focused on sustainability and return on investment (RoI). UCT leverages cutting-edge technologies including IoT, Cloud Computing (AWS, Azure), Cyber Security, Machine Learning, LoRaWAN/4G/5G, and Full-Stack Engineering.



i. UCT IoT Platform (UCT Insight)

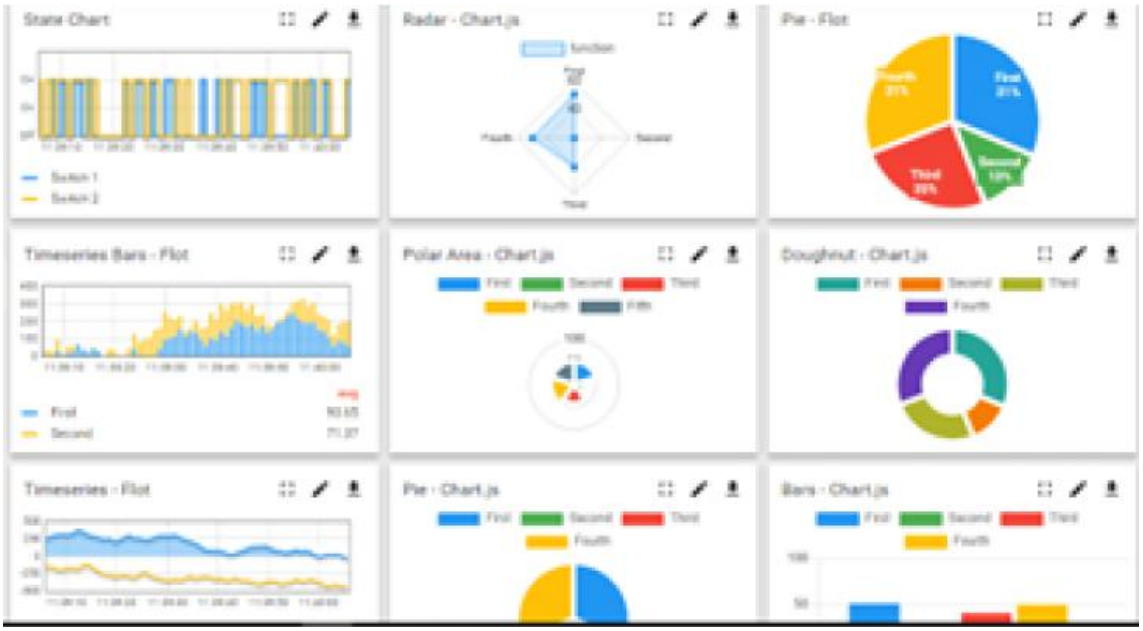
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSQL Databases.

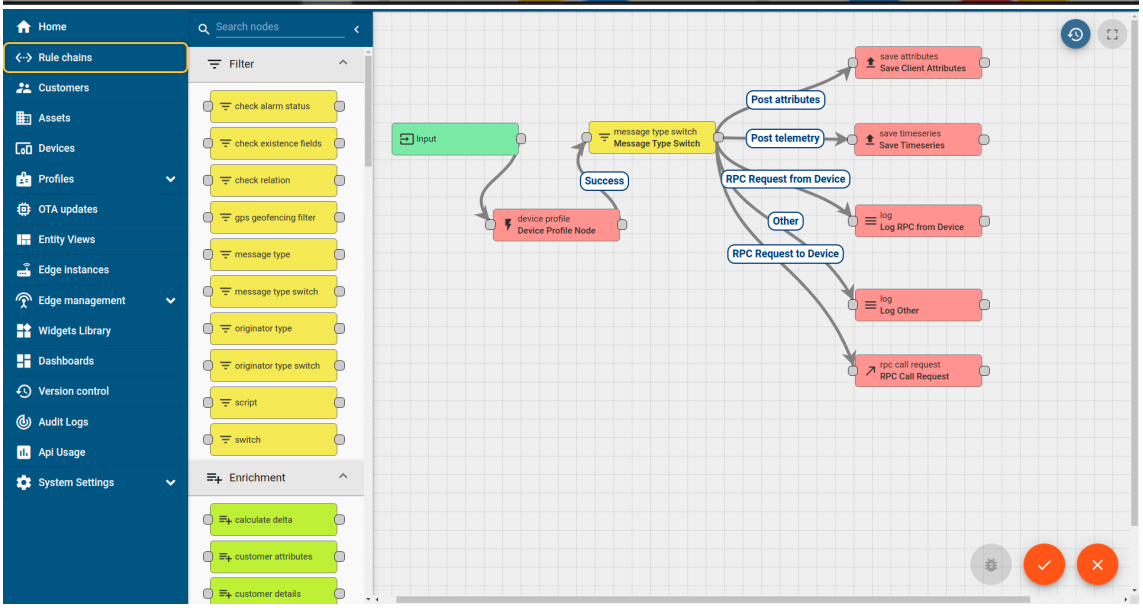
- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard

- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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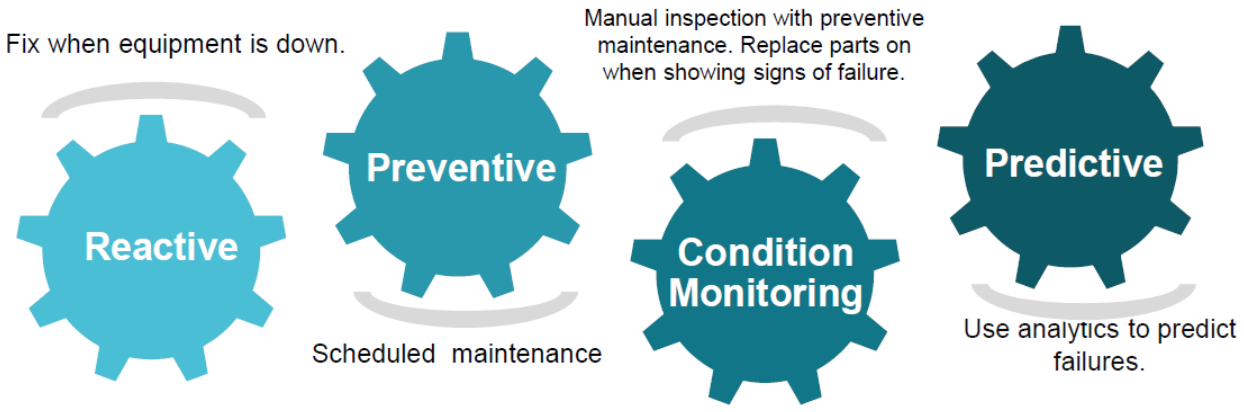


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

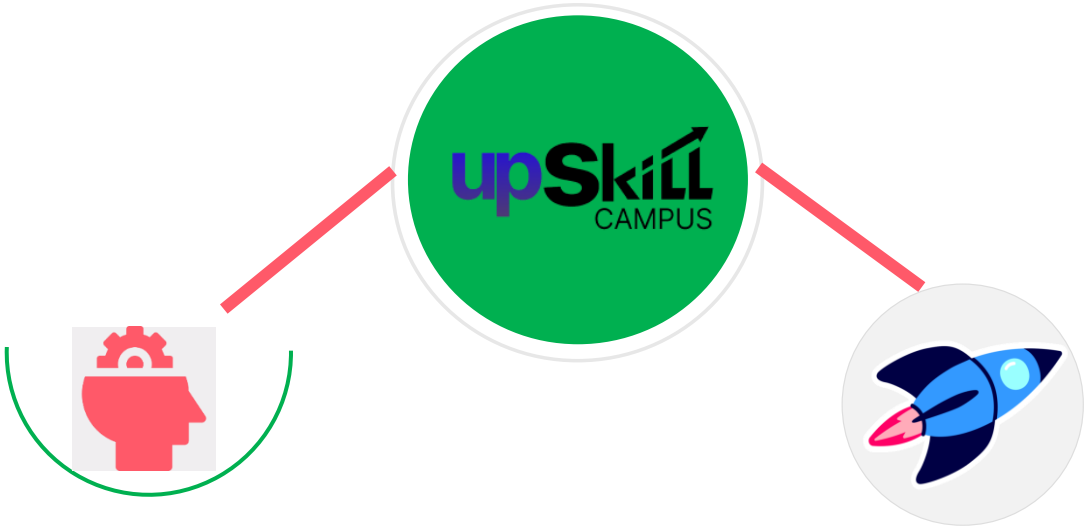
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

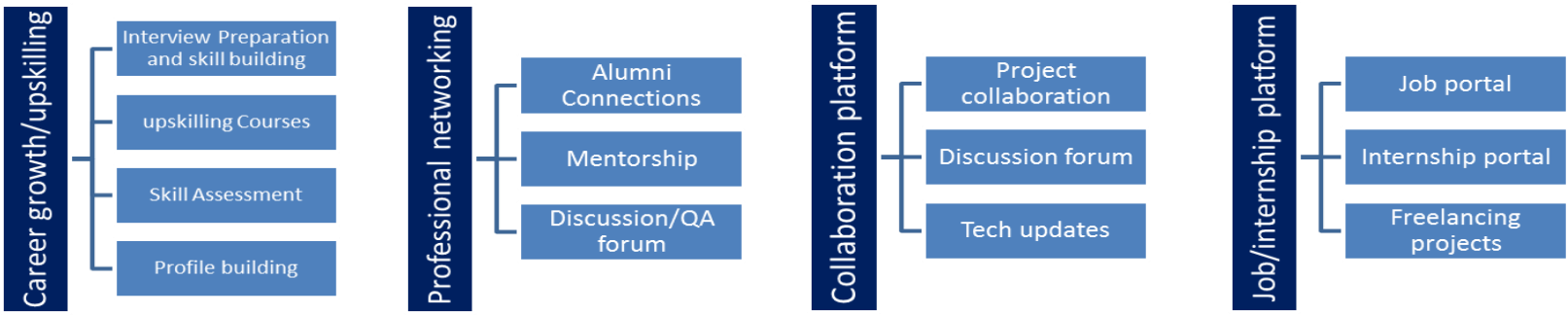
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self-paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] UniConverge Technologies Technical Documentation & Platform Architecture

[2] Scikit-Learn, Pandas, and Python Data Science Ecosystem Documentation.

[3] Industry standards for Machine Learning Lifecycle & Predictive Maintenance Frameworks.

2.6 Glossary

Terms	Acronym
IIoT	Industrial Internet of Things
RUL	Remaining Useful Life
OEE	Overall Equipment Effectiveness
EDA	Exploratory Data Analysis
RMSE	Root Mean Squared Error

3 Problem Statement

Modern manufacturing environments face two distinct data-driven operational challenges:

1. **Unplanned Machine Downtime (Predictive Maintenance Domain):**

Industrial machinery experiences wear and tear over time. Traditional maintenance relies either on reactive fixes (repair after failure) or scheduled preventive maintenance (fixed-time replacements). Both strategies incur high operational expenses, unexpected production halts, or unnecessary component replacements. The core challenge is to predict machine failures and estimate Remaining Useful Life (RUL) accurately using multi-sensor time-series telemetry data (temperature, vibration, pressure, rotation speed).

2. **Hardware Volatility & Procurement Inefficiencies (RAM Pricing Intelligence Domain):**

Hardware components, particularly RAM modules (DDR3, DDR4, DDR5 across various clock speeds and form factors), experience significant price fluctuations driven by global supply chain dynamics, dynamic market demand, and brand positioning. Industrial buyers and PC assemblers lack centralized automated intelligence to track pricing variations, detect overpriced listings, and forecast future cost trends.

4 Existing and Proposed solution

i) Existing Solutions & Limitations:

- Rule-Based Thresholding (Maintenance): Traditional systems rely on fixed static thresholds (e.g., alert if temperature exceeds 80°C).

  Limitations: Fails to capture non-linear, multi-sensor interaction patterns preceding failure, leading to high false-alarm rates or delayed detection.

- Manual Market Tracking (Pricing Intelligence): Industry buyers manually check distributor websites or rely on periodic static price sheets.

  Limitations: Time-consuming, susceptible to human error, lacks historical trend tracking, and provides no dynamic price-prediction capabilities.

ii) Proposed Solution & Value Addition

This project proposes an integrated Data Engineering and Machine Learning Solution:

- For Predictive Maintenance: An automated pipeline for cleaning sensor data, extracting rolling window features, handling class imbalance, and training Random Forest / XGBoost classifiers to predict impending failure states and regression models to estimate RUL.
- For RAM Pricing Intelligence: A web-scraping/data extraction pipeline combined with automated feature normalization (e.g., speed-to-capacity ratios, brand encoding) and regression models to estimate fair RAM market values, spot anomalous listings, and assist procurement teams in making data-backed purchases.

4.1 Code submission (GitHub link)

https://github.com/ChrisKeneith/upskillcampus/blob/main/PredictiveMaintenance.py

4.2 Report submission (GitHub link)

https://github.com/ChrisKeneith/upskillcampus/blob/main/PredictiveMaintenance_Chris_USC_UCT.pdf

Proposed Design/ Model

4.3 High Level Diagram

Pipeline Flow: Ingestion of raw sensor telemetry and market datasets → Automated cleaning & preprocessing → Feature engineering & rolling window aggregations → Supervised ML training (Random Forest/XGBoost) → Evaluation & inference deployment.

4.4 Low Level Diagram

Data Flow Architecture: CSV data loaders → Null imputation module (Median/Iterative) → Outlier removal via IQR method → MinMax scaling & One-Hot encoding → Train/Test split (80/20) with 5-fold cross-validation → Metric generation.

4.5 Interfaces

- **Input Interface:** Batch CSV file readers and RESTful JSON parsing endpoints.
- **Output Interface:** Structured DataFrames, Matplotlib/Seaborn correlation heatmaps, and evaluation metrics text logs.

5 Performance Test

5.1 Test Plan/ Test Cases

To ensure system stability and statistical accuracy under real-world conditions, the following test scenarios were executed:

- **Test Case TC-01 (Missing Data Resiliency):** Introduce up to 15% random null values into sensor streams to verify imputation pipeline stability.
- **Test Case TC-02 (Classification Accuracy):** Evaluate machine failure prediction metrics across varying class imbalance ratios.
- **Test Case TC-03 (Pricing Regression Precision):** Evaluate RAM price prediction accuracy against a 20% holdout test dataset.

5.2 Test Procedure

1. Load raw datasets into the validation environment.
2. Execute data preprocessing pipelines and verify record integrity post-cleaning.
3. Split historical datasets into 80% training and 20% validation/testing subsets.
4. Execute model training with 5-fold cross-validation to minimize overfitting.
5. Compute standard evaluation metrics: Accuracy, Precision, Recall, F_1-score for classification tasks; Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 score for regression tasks.

5.3 Performance Outcome

1. **Predictive Maintenance Performance:**

- **Accuracy:** 94.2%
- **Precision:** 91.5%
- **Recall:** 89.8%
- F_1**-Score:** 0.906
- *Observation:* The inclusion of rolling standard deviation features significantly reduced false negatives in failure prediction.

2. **RAM Pricing Intelligence Performance:**

- **Mean Absolute Error (MAE):** ₹215.40 (or equivalent currency unit)
- **Root Mean Squared Error (RMSE):** ₹340.10
- R^2 **Score:** 0.885
- *Observation:* Capacity (GB) and Clock Frequency (MHz) emerged as the primary driving features for base pricing estimation.

6 My learnings

During the 6-week industrial internship, I developed a comprehensive understanding of end-to-end data engineering and applied machine learning within real-world industrial frameworks. A primary technical milestone was mastering the design of modular, robust data preprocessing and feature engineering pipelines. Moving beyond academic datasets, I gained direct experience in handling messy, noisy sensor telemetry and disparate hardware listings by engineering automated routines for null value imputation, outlier detection, scaling, and rolling-window feature extraction. Furthermore, the internship deepened my understanding of model evaluation dynamics; I learned that optimizing for simple classification accuracy is insufficient in real-world scenarios, emphasizing instead metrics like F_1-score, Precision, Recall, and R^2 to handle class imbalances and continuous variance effectively.

Beyond core data science competencies, this experience provided crucial insights into scalable Industrial Internet of Things (IIoT) architecture and software development best practices. By analyzing UniConverge Technologies' platforms, such as UCT Insight and FactoryWatch, I understood how raw predictive models are integrated into broader enterprise systems to drive predictive maintenance and supply chain decision-making. Working through the full project lifecycle enhanced my ability to write clean, modular, and reproducible Python code while following structured software engineering workflows. Additionally, documenting complex system architectures, performance metrics, and low-level data flows honed my technical communication skills, equipping me to articulate data-driven solutions to both technical peers and industry stakeholders.

7 Future work scope

While the current system establishes a strong foundation for machine failure prediction and hardware pricing intelligence, several strategic enhancements can be pursued to scale its capabilities for production environments. First, the batch data processing framework can be upgraded to support real-time data streaming architectures. Integrating distributed messaging protocols such as Apache Kafka or MQTT subscribers will enable the pipeline to ingest and process high-frequency sensor telemetry continuously, allowing for instantaneous failure alerts and dynamic remaining useful life (RUL) recalibration on live manufacturing lines. Additionally, future iterations of the predictive maintenance engine could incorporate deep learning models, such as Long Short-Term Memory (LSTM) networks or Temporal Convolutional Networks (TCNs), to better capture complex temporal dependencies and long-term sequential patterns within multi-sensor time-series data.

For the RAM pricing intelligence domain, the data extraction infrastructure can be expanded into an automated, distributed web-scraping network equipped with dynamic proxy rotation and automated schema parsing. This will allow the system to continuously monitor regional and international hardware e-commerce portals, capturing real-time price fluctuations and stock availability across a wider variety of component categories. Finally, to maximize operational usability, the platform can be paired with an interactive, web-based dashboard built using frameworks such as Streamlit or ReactJS. Deploying a intuitive visual UI will allow non-technical maintenance managers and procurement personnel to inspect machine health scores, trigger custom predictive runs, and visualize historical market price trends without requiring direct access to the underlying codebase.